Executive Summary: HEARTLY – The Future of Preventative Cardiology

Problem Statement Cardiovascular Diseases (CVD) remain the primary cause of global mortality, yet roughly 80% of cases are preventable through lifestyle modification and early detection. Existing fitness trackers focus on passive monitoring and reactive alerts, failing to provide the predictive, actionable insights necessary to stop a heart event before it occurs.

Current State of the Database HEARTLY utilizes a multi-source data integration strategy to transition users from passive tracking to active risk management. Our current database architecture manages the following entities:

- User Profiles: Demographic information and historical medical records.
- Real-time Metrics: Continuous inputs regarding sleep, diet, exercise, and habits.
- Public Health Datasets: Large-scale disease data used to train our AI risk models.
- Current Services: We provide a dynamic "Heart Score," predictive risk modeling, and hyper-customized lifestyle planning.

PostgreSQL Implementation To ensure high-performance data integrity, we utilize PostgreSQL as our primary relational database. This implementation is critical for:

- Relational Mapping: Managing complex relationships between historical medical records and real-time habit logs.
- Data Security: Leveraging PostgreSQL's robust encryption and role-based access controls to maintain ethical data governance and user anonymity.
- Scalability: Using indexed queries to process multi-source data fusion quickly, ensuring the "Heart Score" remains a real-time metric.

Next Phase: Strategic Expansion. To secure a dominant market position and attract further investment, the next phase focuses on expanding our data entities to enable B2B revenue streams and clinical-grade accuracy:

- New Data Entities: * Genetic Markers: Integrating DNA data to identify hereditary predispositions.
    - Environmental Data: Tracking local air quality and stress-inducing environmental factors via geolocation.
    - Pharmacological Logs: Tracking medication adherence and its real-time impact on cardiac metrics.

By evolving from a consumer app to a comprehensive health ecosystem, HEARTLY will redefine the standard for preventative care.